ANALYSIS REPORT

CVP 1.1 Curve-First Reranking Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report reranks the already accepted `TE Curve Verification Pipeline` candidate matrix by full-curve validation behavior. It does not execute training, does not alter the dataset structure, and does not provide future curve samples to any model.

- Run Instance: `2026-05-28-19-27-46__track2b_curve_first_reranking`
- Source TE Curve Verification Pipeline Run: `output\validation_checks\track2_reference_comparison\2026-05-28-12-22-56__track2_full_directional_family_matrix_wave2c_residual_harmonic_temporal_hybrid_track2_refresh_2026_05_28`
- Source Curve Count: `194`
- Source Candidate Count: `111`
- Generated Artifact Directory: `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking`

Method

The primary ordering key is mean `TE Curve Verification Pipeline` mean-percentage-error over each candidate's valid direction surface. Ties are resolved by P95 mean-percentage-error, worst mean-percentage-error, and mean curve `MAE`. This keeps scalar pointwise registry metrics separate from curve-following evidence.

Available diagnostics from the existing `TE Curve Verification Pipeline` matrix:

- mean curve `MAE` and `RMSE` per operating condition;
- mean percentage error per operating condition;
- P95, worst-condition, and standard-deviation aggregates across conditions.

Deferred diagnostics requiring a future curve-payload export:

- harmonic amplitude and phase error by order;
- derivative or slope continuity error;
- per-revolution residual drift and continuity checks across stitched curves.

Causal Input Boundary

The validation surface is full-curve because the compensation target is continuous `TE` over many consecutive motor revolutions. The runtime input contract remains causal: current point-level operating state, optional short history of already observed samples, or derived causal features only.

Overall Curve-First Leaders

Rank	Candidate	Family	Surface	Direction	Curves	Mean MPE [%]	P95 MPE [%]	Worst MPE [%]
1	rcim_retuned_GBM19_Fw	GBM	Fw	all_valid	97	2.371752	4.911649	5.680916
2	rcim_original_ERT19_Fw	ERT	Fw	all_valid	97	3.252972	6.145160	7.841177
3	rcim_retuned_RF19_Fw	RF	Fw	all_valid	97	3.291549	5.997536	7.048587
4	rcim_original_RF19_Fw	RF	Fw	all_valid	97	3.939860	6.872174	9.475785
5	rcim_original_LGBM19_Fw	LGBM	Fw	all_valid	97	4.016685	10.054446	11.877265
6	rcim_retuned_ERT19_Fw	ERT	Fw	all_valid	97	4.038649	7.599022	9.427596
7	paper_retuned_best_Fw	best_composite	Fw	all_valid	97	4.108527	9.865680	11.737393
8	rcim_retuned_HGBM19_Fw	HGBM	Fw	all_valid	97	4.126307	9.401102	13.400433
9	rcim_retuned_LGBM19_Fw	LGBM	Fw	all_valid	97	4.135160	9.865585	11.737457
10	rcim_retuned_DT19_Fw	DT	Fw	all_valid	97	4.305731	9.062969	12.531740
11	rcim_original_DT19_Fw	DT	Fw	all_valid	97	4.305735	9.062969	12.531740
12	rcim_original_GBM19_Fw	GBM	Fw	all_valid	97	4.312144	8.193384	10.648506
13	rcim_retuned_ET19_Fw	ET	Fw	all_valid	97	4.425530	9.533169	12.508938
14	rcim_original_HGBM19_Fw	HGBM	Fw	all_valid	97	4.493305	10.616761	14.461187
15	rcim_retuned_XGBM19_Fw	XGBM	Fw	all_valid	97	4.587646	10.487803	12.008064

Forward Curve-First Leaders

Rank	Candidate	Family	Surface	Direction	Curves	Mean MPE [%]	P95 MPE [%]	Worst MPE [%]
1	rcim_retuned_GBM19_Fw	GBM	Fw	forward	97	2.371752	4.911649	5.680916
2	rcim_original_ERT19_Fw	ERT	Fw	forward	97	3.252972	6.145160	7.841177
3	rcim_retuned_RF19_Fw	RF	Fw	forward	97	3.291549	5.997536	7.048587
4	rcim_original_RF19_Fw	RF	Fw	forward	97	3.939860	6.872174	9.475785
5	rcim_original_LGBM19_Fw	LGBM	Fw	forward	97	4.016685	10.054446	11.877265
6	rcim_retuned_ERT19_Fw	ERT	Fw	forward	97	4.038649	7.599022	9.427596
7	paper_retuned_best_Fw	best_composite	Fw	forward	97	4.108527	9.865680	11.737393
8	rcim_retuned_HGBM19_Fw	HGBM	Fw	forward	97	4.126307	9.401102	13.400433
9	rcim_retuned_LGBM19_Fw	LGBM	Fw	forward	97	4.135160	9.865585	11.737457
10	rcim_retuned_DT19_Fw	DT	Fw	forward	97	4.305731	9.062969	12.531740
11	rcim_original_DT19_Fw	DT	Fw	forward	97	4.305735	9.062969	12.531740
12	rcim_original_GBM19_Fw	GBM	Fw	forward	97	4.312144	8.193384	10.648506

Backward Curve-First Leaders

Rank	Candidate	Family	Surface	Direction	Curves	Mean MPE [%]	P95 MPE [%]	Worst MPE [%]
1	`rcim_retuned_GBM19_Bw`	`GBM`	Bw	backward	97	5.398275	12.280348	29.298779
2	`periodic_gru_sequence_Bw`	`periodic_gru_sequence`	Bw	backward	97	5.465946	14.820350	18.759599
3	`periodic_gru_sequence_global`	`periodic_gru_sequence`	global	backward	97	6.010388	12.692963	17.220297
4	`periodic_lstm_sequence_Bw`	`periodic_lstm_sequence`	Bw	backward	97	6.013045	15.382036	17.697190
5	`periodic_lstm_sequence_global`	`periodic_lstm_sequence`	global	backward	97	6.097733	14.674261	17.806152
6	`rcim_retuned_ET19_Bw`	`ET`	Bw	backward	97	7.021055	15.287358	28.924505
7	`tree_Bw`	`tree`	Bw	backward	97	7.051484	14.115584	15.808625
8	`tree_global`	`tree`	global	backward	97	7.118450	13.703254	15.694393
9	`rcim_retuned_ERT19_Bw`	`ERT`	Bw	backward	97	7.269106	13.186813	29.804962
10	`residual_harmonic_lstm_sequence_sparse_rcim_Bw`	`residual_harmonic_lstm_sequence_sparse_rcim`	Bw	backward	97	7.510266	13.159550	15.012015
11	`rcim_retuned_RF19_Bw`	`RF`	Bw	backward	97	7.542522	15.083100	28.325982
12	`paper_retuned_best_Bw`	`best_composite`	Bw	backward	97	7.571630	15.644903	29.854827

Surface Leaders

Surface	Leader	Family	Source	Curves	Mean MPE [%]	P95 MPE [%]	Mean Curve MAE [deg]
Bw	`rcim_retuned_GBM19_Bw`	`GBM`	rcim_retuned	97	5.398275	12.280348	0.002766
Fw	`rcim_retuned_GBM19_Fw`	`GBM`	rcim_retuned	97	2.371752	4.911649	0.001089
global	`periodic_lstm_sequence_global`	`periodic_lstm_sequence`	wave2_temporal_entry_registry	194	6.119950	14.716986	0.002707

Scalar Registry Context

- Current scalar registry winner: `te_periodic_gru_sequence_remote_Bw` from family `periodic_gru_sequence_bw`.
- Scalar test `MAE` : `0.002344` and scalar test `RMSE` : `0.002747`.

  The strongest forward curve-first candidate in this reranking is `rcim_retuned_GBM19_Fw` from family `GBM` with mean `MPE` `2.371752` percent and P95 `MPE` `4.911649` percent. This does not replace the backward or global branches.

Decision

This pass standardizes the curve-first evidence surface and should be read as three parallel selection tracks: `Fw`, `Bw`, and `global`. It does not promote one single program-best model by itself because the real application needs one best candidate per surface and richer harmonic/phase diagnostics still require curve-payload export.

Machine-readable artifacts

- `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking\candidate_curve_first_ranking.csv`
- `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking\direction_curve_first_ranking.csv`
- `output\validation_checks\track2_curve_first_reranking\2026-05-28-19-27-46__track2b_curve_first_reranking\track2_curve_first_reranking_summary.yaml`